

NEURAL BELLMAN OPERATORS

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We study when a computed policy can be reused after an economic contract changes. Neural Bellman Operators separate policy proposal, evaluation, improvement, and certification. For a stopped portfolio economy with adjustable risk aversion, operating benefits and liquidation levels affect behavior through a single contract contrast. Value envelopes certify welfare losses uniformly across a continuum of contracts; action-separation bounds distinguish portfolio reversals from numerical ambiguity. The implementation repairs improvable neural proposals and evaluates all controls, stopping times, and adjustment effort. A coupled resource-allocation experiment isolates the contributions of neural continuation values and actor initialization, with cost-matched comparisons at increasing query volumes. Persistent capacity competition, recursive utility, and temporal selves provide further tests. The analysis connects operator errors to economic decisions without equating a small training residual with an optimal policy.

KEYWORDS: Dynamic programming, Neural policy iteration, Recursive utility, Endogenous preferences, Stochastic control, Computational economics.

1. INTRODUCTION

Economic counterfactuals rarely require only one solution of a dynamic problem. A researcher changes an operating benefit, a cost of adjustment, or an outside option and asks whether the policy response is economically significant. Repeating a numerical solve at each parameter value can be costly. Reusing an approximate policy without knowing its value loss can instead confound an economic reversal with approximation error. This problem is especially acute when controls affect both financial resources and the agent’s preferences, and when their joint state determines the termination of an operating contract.

This paper develops Neural Bellman Operators (NBO) as a framework for policy computation and contract counterfactuals. The computational objects are separated: a critic approximates continuation value; an actor proposes a feasible control; an improvement calculation checks that proposal; and an independent evaluation measures the induced policy. Neural approximation is useful only insofar as its errors and computational costs can be related to the economic decision being studied. We therefore make the policy-value guarantee, rather than a neural training objective, the central object of the analysis.

The main application is portfolio choice with adjustable risk aversion and first-exit liquidation. It retains consumption, risky investment, costly preference adjustment, two stochastic shocks, and their covariance. The liquidation payoff is an economic primitive, not a numerical transfer that returns wealth to the interior. We show that a constant operating benefit m and a constant liquidation increment ℓ influence choices through the single contrast $d = m - \rho\ell$. The separate levels are observationally equivalent along an annuity-normalization direction. Changing m alone changes the contract, whereas

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shifting both components coherently cannot reverse the portfolio. This distinction separates an operating-duration incentive from an unconditional preference-hedging interpretation.

We then construct a certified policy library over the contract contrast. Independent evaluation supplies each stored policy’s baseline benefit, discounted duration, and adjustment effort. Feasible policy values form a lower envelope; certified anchor values form an upper envelope. Statewise switching among stored policies dominates their lower envelope. A finite breakpoint calculation controls the value loss at every state and date for every parameter in an interval, not only at sampled parameter values. A second calculation bounds the value of all opposite-sign first actions. It certifies economic decision regions even when a global welfare tolerance alone would be too coarse to identify a portfolio sign.

The executed experiment uses the positive finite approximation of the stopped economy and includes both the reference and diagnostic action meshes together with the frozen neural proposals. It repairs the substantial feasible deviations of the initial neural fits by using evaluated policy continuations. The resulting library attains a uniform 10^{-3} welfare target over the entire displayed contract interval. The initial portfolio is provably negative over one subinterval and positive over another, with an explicitly reported unresolved band between them. Joint changes in time, state, and action resolution retain the endpoint reversal, while their variation near the switch remains visible. These are quantitative conclusions about specified finite economies; they are not inferred from a small differential residual or presented as a Brownian first-passage error estimate.

A coupled stochastic resource-allocation problem provides a separate test of representation and computational reuse. Its shared investment budget binds, shocks are correlated, and the state cost is nonseparable. An independent nonanticipative scenario-tree problem bounds each tested policy’s lifetime cost loss. Same-checkpoint deployment comparisons isolate a neural actor’s role as an initialization device. A full convex-quadratic continuation fit tests whether a simpler representation attains the same accuracy. Fresh-state query workloads charge for critic training, online improvement, and complete policy evaluation, and compare them with a solver exploiting the model’s convex structure. This identifies the workload at which reuse is useful without attributing the entire advantage to the actor.

The general error analysis and the broader economic extensions remain part of the argument. A finite-horizon theorem distinguishes evaluation, improvement, boundary, and transition-operator errors. Exact policy improvement supplies a conditional benchmark, while counterexamples explain why an attracting actor stationary point or a small smooth residual is insufficient. Merton and Epstein–Zin problems test actual evaluation and control updates. Sophisticated temporal selves require a different improvement criterion from ordinary continuation evaluation. A persistent-capacity game tests full unilateral dynamic deviations at every date and state. These exercises serve different purposes and are not combined into a single success count.

1.1. *Related Literature and the Contribution*

Policy improvement and approximate dynamic programming precede neural approximation. [Jacka and Mijatović \(2015\)](#) study continuous-time policy improvement under explicit control hypotheses. Our exact-operator proposition records the boundary and limiting requirements needed for that benchmark. The finite-horizon error account is

backward propagation through monotone operators; the empirical contribution is to make the relevant error information operational for an economic counterfactual.

The nearest neural approaches also combine value approximation and control improvement. Kim, Kim, Kim, and Cho (2025) use neural fixed-policy PDE evaluation within policy iteration; Kim, Kim, Kim, and Cho (2026) develop interior error bounds when boundary information is unresolved. Lee and Kim (2024) combine policy iteration with deep operator learning, including transfer across terminal functions. Cai, Fang, and Zhou (2025) use martingale conditions for value and control representations. Direct neural approximation of stochastic controls appears in Han and E (2016). Neural evaluation, separate parameter blocks, and avoidance of a closed-form argmax are consequently not claims of priority here.

Our policy library is closely related to successor features and generalized policy improvement (Barreto et al., 2017), policy switching (Chang, 2021), and policy caches (Nemecek and Parr, 2021). In particular, the convex anchor upper bound is already available in the policy-cache literature. We use it, rather than present it as a new discovery. The contribution developed here is an economic and computational specialization with three jointly executed requirements: an exact reduction of stopped-contract primitives; a certified, adaptively constructed continuum of counterfactuals using independently evaluated policies; and action-separation bounds that distinguish a portfolio reversal from numerical uncertainty. The application also reports actual repair costs and complete-query workloads, including the simpler alternatives that make the interpretation of neural approximation testable.

The broader computational literature includes projection and recursive methods (Judd, 1998, Stokey, Lucas, and Prescott, 1989), adaptive sparse grids (Brumm and Scheidegger, 2017), and neural differential-equation solvers (Raissi, Perdikaris, and Karniadakis, 2019, Han, Jentzen, and E, 2018). The appropriate comparator exploits the problem’s structure. We use Riccati recursion for an LQR benchmark and convex optimization for the constrained resource problem. The convex neural critic is an application of input-convex approximation (Amos, Xu, and Kolter, 2017), not a new architecture. Its role is to represent a continuation function more flexibly than a convex quadratic while retaining a checkable one-period improvement gap.

The following sections define the operators, establish the approximation account, specify the preference economy, develop contract reuse and action certification, and report the computations. The final substantive section treats recursive utility, temporal selves, and capacity competition. The standalone supplement retains the operator proofs, negative controls, equilibrium conditions, and replication details and adds the contract and all-date verification calculations.

2. ECONOMIC CONTROL AND SEPARATED BELLMAN OPERATORS

2.1. The Controlled Problem

Let $S_t = (u_t, X_t)$ be a Markov state and let $a_t = (\theta_t, b_t)$ collect internal and external controls. On a specified domain D , the controlled state satisfies

$$dS_t = \mu(t, S_t, a_t) dt + B(t, S_t, a_t) dW_t, \quad a_t \in A(t, S_t), \quad (1)$$

where W is a vector of independent Brownian motions and $A(t, s)$ is the feasible correspondence. Correlation is represented in B . More generally, a continuous martingale with covariance density $Q(t, s, a)$ gives covariance $\sigma Q \sigma^\top$. A predictable covariance

process not determined by the declared state does not, merely by notation, define the same Markov problem; the state must be augmented or the formulation changed.

For additive preferences and stopping time $\tau \leq T$, policy evaluation is

$$J^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(v-t)} r(v, S_v, \pi_v) dv + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right]. \quad (2)$$

The stopping rule and payoff are economic primitives. In an exit problem, τ is the first exit or terminal date. Reflection instead requires a regulator and a boundary operator involving its direction; state constraints require viability. Neither interpretation is supplied by an unreported numerical projection.

For recursive preferences, evaluation is first defined for each admissible policy by

$$-dY_t^\pi = f(t, S_t, \pi_t, Y_t^\pi) dt - Z_t^\pi dW_t, \quad Y_\tau^\pi = g(\tau, S_\tau). \quad (3)$$

The driver may incorporate a preference-adjustment cost. Existence, integrability, comparison, and any domain restriction on its value argument are imposed before taking a supremum over policies. Lipschitz drivers on a finite horizon provide a standard admissible class; the Epstein–Zin specification below uses its own positive-domain and transversality conditions. Calling Y an argument of utility does not itself establish well-posedness.

When a classical Markov representation is available, write

$$\mathcal{H}^a[v] = f(t, s, a, v) + \mu(t, s, a)^\top \nabla v + \frac{1}{2} \text{tr}\{B(t, s, a)B(t, s, a)^\top \nabla^2 v\}. \quad (4)$$

For additive utility, $f = r - \rho v$. The evaluation equation is $v_t + \mathcal{H}^\pi[v] = 0$ with the specified boundary operator, while the control equation replaces $\mathcal{H}^\pi$ by $\sup_{a \in A} \mathcal{H}^a$. Boundary data, admissibility, and comparison are part of these statements, not consequences of a neural parameterization.

2.2. Evaluation, Proposal, and Improvement

Let v_ξ be a critic and a_ω a feasible actor. A differential evaluation step freezes the policy and minimizes

$$L_E(\xi; \pi) = \mathbb{E}_v \left[(\partial_t v_\xi + \mathcal{H}^\pi[v_\xi])^2 \right] + \lambda_B \mathbb{E}_{v_B} \left[|\mathcal{B}v_\xi - g_B|^2 \right]. \quad (5)$$

Here $\mathcal{B}$ is the actual boundary operator. A Dirichlet discrepancy is appropriate for the stopped problem used below, not for an unspecified reflected diffusion. Hard architectures can impose the boundary exactly, although they do not eliminate interior approximation error.

The actor’s Hamiltonian proposal solves, approximately,

$$\max_{\omega} \mathbb{E}_{v_I} \left[\mathcal{H}^{a_\omega} \{ \text{sg}(v_\xi), \text{sg}(\nabla v_\xi), \text{sg}(\nabla^2 v_\xi) \} \right]. \quad (6)$$

The notation sg freezes the critic’s parameter block. Coefficients depending on the action, including diffusion loadings, retain their action derivatives. Evaluation does not update actor parameters, and actor improvement does not update critic parameters. For a game, rival controls are also frozen in a player’s improvement step. These restrictions concern the computational graph, not just the labels attached to two loss functions.

A bounded activation can enforce a box constraint. Shared resource constraints require a different map. For example, $a_i = B_0 \exp(z_i) / (\sum_{j=1}^d \exp(z_j) + \exp(z_0))$ gives $a_i \geq 0$ and $\sum_i a_i \leq B_0$, with z_0 representing slack. General convex sets can use projection; a penalty alone is an approximation whose violations must be measured. An arbitrary policy sampling measure does not turn (6) into the gradient of lifetime welfare. Such an interpretation requires occupancy or adjoint weighting and the relevant interchange conditions.

For an implemented transition rule, it is often preferable to use a finite-step operator. At dates $n = 0, \dots, N - 1$, define

$$(\mathcal{T}_n^a v)(s) = R_n(s, a) + \int v(s') K_n(ds' | s, a), \quad \mathcal{T}_n v = \sup_{a \in A_n(s)} \mathcal{T}_n^a v. \quad (7)$$

The nonnegative kernel includes discount and survival. A stopped payoff is included in R_n ; it is not continued at a clipped interior state. The experiments with nonlinear critics proceed backward: improve the actor against the frozen next-date critic, then fit the current critic to the evaluated target of the selected action. This is a finite-horizon separated Bellman update, not a simulation of a joint scalar loss.

In this semigroup implementation, freezing the continuation network's parameters must not remove its derivative with respect to the next state. The actor influences s' through the transition, so the composition $v_{n+1}(F(s, a, \varepsilon))$ must retain that input derivative. Our tests compare this actor gradient with finite differences and separately check the differential Hamiltonian graph. The distinction prevents a stop-gradient convention appropriate for a Hamiltonian jet from being incorrectly imposed on a finite-step transition.

2.3. The Exact Policy-Improvement Benchmark

The following proposition states the exact object to which neural approximation is compared. Let $\mathcal{V}$ be a classical value space whose norm controls $v, v_t, \nabla v, \nabla^2 v$, and the boundary traces required by $\mathcal{B}$. Including the time derivative is essential to pass the evaluation equation to a limit.

ASSUMPTION 1—Exact evaluation and improvement: *The policy class Π is compact in a stated uniform topology. Evaluation $E : \Pi \rightarrow \mathcal{V}$ is uniquely defined, continuous, and has precompact image. A selector I attains the pointwise Hamiltonian supremum, maps Π into itself, and is continuous at accumulation points of the policy sequence. Comparison and verification hold for the specified generator, recursive driver, boundary operator, and value class. Evaluated values are bounded above by the optimal value.*

PROPOSITION 1—Exact separated policy improvement: *Under Assumption 1, let $\pi_{j+1} = I(\pi_j)$ and $v_j = E(\pi_j)$. The evaluated values increase to a common limit in $\mathcal{V}$. Every accumulation policy p satisfies $E(p) = E(I(p)) = \bar{v}$, and $\bar{v}$ solves the maximized boundary-value problem. If comparison identifies the value, p is optimal.*

The important limit is equality of values, not equality of successive policies. Along $\pi_{j_k} \rightarrow p$, continuity gives $E(p) = \bar{v}$ and $E(I(p)) = \bar{v}$ from the shifted sequence. The evaluation equation for $I(p)$ at that same value is the maximized equation. The supplement proves this step in detail. Compactness and closure under an exact selector are substantial hypotheses; they are not automatically properties of a finite neural class. Proposition 1 is not an optimizer convergence theorem.

3. APPROXIMATION ERROR AND FEASIBLE IMPROVEMENT

3.1. An Error Account for Policy Value

Let $\mathcal{T}_n^a$ be the target evaluation operator and $\tilde{\mathcal{T}}_n^a$ the implemented operator. For additive preferences, these have the form (7). For recursive preferences, the following result also permits nonlinear evaluation maps, provided they are order preserving and Lipschitz in the continuation value. We impose those properties on the relevant value domain rather than presume them for every recursive aggregator.

ASSUMPTION 2—Finite-horizon operators: *For every feasible a , the target operator $\mathcal{T}_n^a$ is order preserving and satisfies $\|\mathcal{T}_n^a v - \mathcal{T}_n^a w\|_\infty \leq \gamma_n \|v - w\|_\infty$. The same bound holds for $\mathcal{T}_n = \sup_a \mathcal{T}_n^a$; the supremum is well defined. Terminal values and the selected policy values exist and are bounded. The numerical errors defined below are finite. The constants γ_n may exceed one for a finite-horizon recursive problem.*

For learned values $\hat{v}_n$ and a feasible policy π_n , define

$$\epsilon_n = \left\| \hat{v}_n - \tilde{\mathcal{T}}_n^{\pi_n} \hat{v}_{n+1} \right\|_\infty, \quad (8)$$

$$\delta_n = \left\| \tilde{\mathcal{T}}_n \hat{v}_{n+1} - \tilde{\mathcal{T}}_n^{\pi_n} \hat{v}_{n+1} \right\|_\infty, \quad (9)$$

$$\kappa_n = \sup_{s,a} \left| \tilde{\mathcal{T}}_n^a \hat{v}_{n+1}(s) - \mathcal{T}_n^a \hat{v}_{n+1}(s) \right|, \quad \eta = \|\hat{v}_N - g_N\|_\infty. \quad (10)$$

The first quantity is evaluation error; the second is a payoff improvement gap, not an action distance. The third compares operators on the actual learned continuation. It includes transition, quadrature, interpolation, and stopped-payoff errors when those are present. Let $\Gamma_{n,j} = \prod_{i=n}^{j-1} \gamma_i$, with an empty product equal to one.

THEOREM 2—Policy-value error account: *Under Assumption 2, the target policy value J_n^π satisfies*

$$\|\hat{v}_n - J_n^\pi\|_\infty \leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \kappa_j) + \Gamma_{n,N} \eta. \quad (11)$$

For the target optimal value V_n^* ,

$$0 \leq V_n^*(s) - J_n^\pi(s) \leq \sum_{j=n}^{N-1} \Gamma_{n,j} (2\epsilon_j + \delta_j + 2\kappa_j) + 2\Gamma_{n,N} \eta. \quad (12)$$

The proof is backward error propagation, but the placement of the errors matters. The policy-evaluation recursion contributes $\epsilon_n + \kappa_n$. The optimized recursion contributes $\epsilon_n + \delta_n + \kappa_n$. Adding the resulting bounds yields (12); separately inflating an actor gap by operator error would produce an unnecessarily loose coefficient. For additive stopped control with exact discount, $\gamma_n \leq e^{-\rho \Delta_n}$. A boundary-payoff error at an intermediate exit belongs in the relevant one-step operator error, not only in the terminal error at N .

There is an analogous differential statement. If a classical candidate v has boundary discrepancy at most η , fixed-policy residual at most $\epsilon(t)$, and Hamiltonian gap at most $\delta(t)$, Itô's formula and verification give

$$V^*(t, s) - J^\pi(t, s) \leq 2\eta + \int_t^T e^{-\rho(q-t)} \{2\epsilon(q) + \delta(q)\} dq. \quad (13)$$

The stopped martingale and comparison conditions are stated in Supplement S.1. This is a quantitative link to a differential critic, not a claim that sampled residuals provide the required supremum bounds.

A finite test set by itself provides a lower observation of a supremum, not an upper certificate. With a known Lipschitz constant L and an h -cover, a residual's uniform magnitude is bounded by its largest tested magnitude plus Lh . Without such regularity and coverage information, we report held-out errors as diagnostics. The same distinction applies to critic derivatives, boundary layers, and off-policy states. Network approximation of a smooth function does not convert a sample average into a uniform error bound.

3.2. Two Improvement Safeguards

Let $Q_n(s, a) = \tilde{T}_n^a \hat{v}_{n+1}(s)$. For a finite feasible set $C_n(s)$ and neural proposal $a_\omega(s)$, the candidate safeguard selects

$$\pi_n(s) \in \arg \max_{a \in C_n(s) \cup \{a_\omega(s)\}} Q_n(s, a). \quad (14)$$

It weakly improves the proposal's evaluated target at every queried state. If $C_n(s)$ is an h_a -cover of the feasible set and Q_n has a verified action-Lipschitz constant L_a , its remaining improvement gap is at most $L_a h_a$. Without that information, a finite comparison is not a continuum optimality certificate. In the portfolio computation we use the safeguard and report the remaining independently evaluated feasible deviations.

For convex cost minimization, a stronger check is available. Let $F_s(a)$ be differentiable and convex on a compact convex feasible set $A(s)$. Define

$$g_s(a) = \nabla F_s(a)^\top a - \min_{b \in A(s)} \nabla F_s(a)^\top b. \quad (15)$$

Convexity implies $0 \leq F_s(a) - \min_b F_s(b) \leq g_s(a)$. For $A = \{a \geq 0 : \sum_i a_i \leq B_0\}$, the linear minimum is $B_0 \min\{0, \min_i \partial_i F_s(a)\}$. The computation therefore obtains an improvement certificate without a global search over neural parameters. The actor is a proposal; projected improvement is stopped according to (15).

In the resource application, the continuation critic is convex by construction, so the one-period criterion is strongly convex in the action. Its neural representation uses nonnegative coefficients on squared-ReLU features together with a convex quadratic term and a free affine term. This architecture justifies the convex check for the *learned* continuation criterion. It does not assume that its continuation value is exact. An independent scenario-tree calculation separately assesses the resulting policy's lifetime cost.

PROPOSITION 3—Finite stochastic-program policy certificate: *Let $C(u)$ be the convex cost of a nonanticipative finite scenario-tree control problem on a compact convex feasible set $\mathcal{U}$. Let $\hat{u}$ be a feasible reference point, $G(\hat{u})$ its gap defined as in (15), and u^π the feasible tree induced by an evaluated feedback policy. Then*

$$C(u^\pi) - C(\hat{u}) \leq C(u^\pi) - C^* \leq C(u^\pi) - C(\hat{u}) + G(\hat{u}). \quad (16)$$

This certificate is independent of the training residual. It applies at each initial state whose complete scenario tree is evaluated. It is not a bound for an untested continuum of initial conditions, nor does it cover a different shock distribution. The supplement gives the projected-simplex implementation and the independent tree gradient.

3.3. Stationarity and Nonsmooth Solutions

A smooth Hamiltonian can have a strictly suboptimal attracting stationary point. On $[-2, 2]$, consider

$$H(a) = -(a^2 - 1)^2 + 0.2a. \quad (17)$$

Its negative local maximizer is approximately -0.973994 , while its global maximizer is approximately 1.024120 . The former is locally attracting under gradient ascent and has an improvement gap of approximately 0.399875 . Exact evaluation of $V^a(t) = (T - t)H(a)$ does not change that fact. Isolation of stationary points cannot turn an actor-ODE limit into an optimal policy. An optimality conclusion needs, for example, a justified global concavity or gradient-dominance condition together with the necessary representation and evaluation assumptions. Constant-step Adam and finite-step L-BFGS are not covered by such a conclusion merely because they update separate parameter blocks.

Nor does a small differential residual select a viscosity solution. For the exit equation $1 - |v'| = 0$ on $(-1, 1)$ with zero boundary values, the relevant value is $|x| - 1$. The functions

$$w_\varepsilon(x) = \sqrt{1 + \varepsilon^2} - \sqrt{x^2 + \varepsilon^2} \quad (18)$$

have the same boundary values and squared residual tending to zero, but converge to the wrong function $1 - |x|$. At zero, that function fails the viscosity subsolution inequality. Our regression suite retains this example and the biased composite-loss example in Supplement S.2. A convergent approximation of a nonsmooth control problem needs an appropriate mechanism, such as monotonicity, stability, consistency, boundary treatment, and comparison (Barles and Souganidis, 1991); it is not supplied by smooth activation functions alone.

3.4. The Squared-Trace Objective

For a differential critic, let $A = B^\top \nabla^2 v B$ and let z_k be independent standard Gaussian probes. Conditional on a fixed state, action, and parameter vector, put $q_k = z_k^\top A z_k$, $\bar{q} = K^{-1} \sum_k q_k$, and $r_k = a - q_k/2$. Then

$$\mathbb{E}(a - \bar{q}/2)^2 = (a - \text{tr}(A)/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (19)$$

Averaging r_k^2 instead has an excess expectation independent of K . These are different objectives, even though each individual trace estimate is unbiased.

PROPOSITION 4—An unbiased squared-residual estimand: For $K \geq 2$, define

$$U_K = \frac{\left(\sum_{k=1}^K r_k\right)^2 - \sum_{k=1}^K r_k^2}{K(K-1)}. \quad (20)$$

Then $\mathbb{E}U_K = (a - \text{tr}(A)/2)^2$. If differentiation and expectation can be interchanged, its parameter gradient is unbiased for the gradient of that squared residual.

The estimator can be negative in a batch. Unbiasedness does not establish low variance, stability of its minimization, or an accurate policy. Optimizing over actions using the same noisy probes also introduces a selection issue beyond the fixed-action identity. We therefore retain exact derivatives for the differential benchmark checks and assess the probe estimands with repeated independent batches. The executed high-dimensional resource solver uses a finite-step operator and a convex critic; its results are not attributed to Hutchinson acceleration.

4. ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

4.1. A Stopped Operating Economy

We now specify an economy in which the preference-adjustment margin is active in the transition law. Wealth is X , risk aversion is u , and the controls are consumption c , preference adjustment θ , and risky-asset share π . Inside $D = (1.2, 2.8) \times (0.5, 2.0)$, the state evolves as

$$du_t = \theta_t dt + \sigma_u dZ_t^u, \quad (21)$$

$$dX_t = \{(r + \pi_t(\mu - r))X_t - c_t\} dt + \pi_t X_t \sigma_X dZ_t^X, \quad d\langle Z^u, Z^X \rangle_t = \varrho dt. \quad (22)$$

The operating contract terminates at $\tau = T \wedge \inf\{t : S_t \notin D\}$. At that time the agent receives the cardinal payoff

$$G(u, X) = -0.02(u - 2)^2 + 0.1 \log X. \quad (23)$$

The payoff summarizes liquidation or the outside option after leaving the operating regime; it is not a transfer restoring the state to the interior.

The feasible box is $c \in [0.05, 0.8]$, $\theta \in [-0.2, 0.2]$, and $\pi \in [-0.5, 0.8]$. We set

$$T = 1, \quad \rho = 0.04, \quad r = 0.02, \quad \mu = 0.08, \quad \sigma_X = 0.2, \quad \sigma_u = 0.05, \quad \varrho = -0.25. \quad (24)$$

The baseline adjustment-cost coefficient is $k = 2$. For a feasible policy,

$$J_k^\pi(t, u, X) = \mathbb{E}_{t,u,X}^\pi \left[\int_t^\tau e^{-\rho(s-t)} \left\{ \frac{c_s^{1-u_s}}{1-u_s} - \frac{k}{2} \theta_s^2 \right\} ds + e^{-\rho(\tau-t)} G(u_\tau, X_\tau) \right]. \quad (25)$$

Consumption is measured in the fixed units appearing in (25). An additive or multiplicative normalization depending on u changes incentives once u is controlled. We therefore do not treat such transformations as innocuous changes of utility representation.

This boundary contract is a substantive specification. Constant preference volatility cannot remain in a closed interval without an exit or regulator mechanism. Moreover,

under the displayed consumption bounds, the drift at $X = 0.5$ is negative even at the largest feasible portfolio share: its maximum is -0.016 . An unregulated state-constrained interpretation is consequently unavailable. Stopping repairs that problem without modifying the interior diffusion, imposing an unreported reflection, or injecting wealth. It is not equivalent to the earlier clipped transition.

The stopped problem has bounded payoffs and Lipschitz coefficients before exit. A useful check is that G is a strict supersolution for every feasible control. The bounds in (24) imply

$$\frac{c^{1-u}}{1-u} - \frac{k\theta^2}{2} + \mathcal{L}^{c,\theta,\pi}G - \rho G \leq -0.8162 \quad (k \geq 0). \quad (26)$$

The supplement derives the bound without a grid maximization. Together with a fixed nondegenerate admissible diffusion, it supplies a boundary-continuity argument for the stopped value. The comparison also has economic content: continued operation has a cost relative to the liquidation payoff. Exit incentives are therefore part of this calibration and must be reported, rather than interpreted as a generic wealth or risk-aversion effect.

4.2. The Operator and Control Margins

Writing subscripts for derivatives, the HJB equation in the interior is

$$\begin{aligned} -V_t = \sup_{c,\theta,\pi} \{ & \frac{c^{1-u}}{1-u} - \frac{1}{2}k\theta^2 - \rho V + \theta V_u + [(r + \pi(\mu - r))X - c]V_X \\ & + \frac{1}{2}\sigma_u^2 V_{uu} + \frac{1}{2}\pi^2 X^2 \sigma_X^2 V_{XX} + \varrho \sigma_u \sigma_X \pi X V_{uX} \}, \end{aligned} \quad (27)$$

with $V = G$ on the spatial boundary and at T . Both diagonal diffusion terms and the mixed term belong to the operator. In particular, the generator of u^2 contains σ_u^2 even if the preference drift is zero.

For an interior optimum, the first-order conditions are

$$c^{-u} = V_X, \quad k\theta = V_u, \quad 0 = (\mu - r)XV_X + \pi X^2 \sigma_X^2 V_{XX} + \varrho \sigma_u \sigma_X X V_{uX}. \quad (28)$$

When $V_X > 0$, $V_{XX} < 0$, and all relevant controls are interior, these give

$$c = V_X^{-1/u}, \quad \theta = V_u/k, \quad \pi = -\frac{(\mu - r)V_X}{X\sigma_X^2 V_{XX}} - \frac{\varrho \sigma_u V_{uX}}{X\sigma_X V_{XX}}. \quad (29)$$

The final expression separates a mean-variance term from a preference-hedging term. Its sign is not determined by $\varrho < 0$ alone: the cross derivative and curvature also matter. Nor need the stopped economy satisfy $V_X > 0$ throughout the domain. The constrained implementation compares feasible controls rather than mechanically applying these formulas where their premises fail. At a binding box constraint, the appropriate one-sided derivative or complementary-slackness condition replaces an interior equality.

The cardinal utility derivative is

$$\partial_u \left(\frac{c^{1-u}}{1-u} \right) = \frac{c^{1-u} \{1 - (1-u) \log c\}}{(1-u)^2}. \quad (30)$$

It need not have the sign suggested by a verbal interpretation of increased risk aversion. The full shadow value V_u incorporates this derivative, future opportunities, adjustment costs, covariance, and liquidation. The numerical policy is informative only relative to these declared primitives.

4.3. A Comparative Static That Survives Endogenous Policies

Fix an initial state and let

$$C^\pi = \mathbb{E}^\pi \int_0^\tau e^{-\rho t} \frac{\theta_t^2}{2} dt, \quad B^\pi = J_k^\pi + kC^\pi. \quad (31)$$

For a fixed policy, B^π and C^π are independent of k : changing the coefficient does not change the state law unless the policy is changed. In particular, the stopping time is allowed to be endogenous to the policy.

THEOREM 5—Adjustment costs and cumulative effort: *Suppose the feasible policy class and all primitives other than k are fixed, and optimizers exist at the coefficients being compared. Then $V(k) = \sup_\pi \{B^\pi - kC^\pi\}$ is decreasing and convex. If $k_2 > k_1$ and π_i is optimal at k_i , then*

$$C^{\pi_2} \leq C^{\pi_1}. \quad (32)$$

Each $-C^{\pi_i}$ is a subgradient of V at k_i . At differentiability points, all optimal effort levels equal $-V'(k)$. For policies with objective losses at most e_1, e_2 at the respective coefficients,

$$C^{\pi_2} - C^{\pi_1} \leq \frac{e_1 + e_2}{k_2 - k_1}. \quad (33)$$

The proof uses the two optimality inequalities and is given in Supplement S.3. The approximate version is useful when numerical policies are used in an economic comparison: the effort ordering can be assessed against a welfare-error budget. The result does not hold pointwise for $\theta(t, u, X)$ merely because (29) contains $1/k$. The shadow value, portfolio choice, consumption, and exit distribution all change when the problem is re-solved.

The theorem is an economic comparative static, not an identification result. Inferring k from data additionally requires an observation model for latent preferences, an identified moment map, and control of numerical error in that map. The solver supplies policy and value objects for such a model; a graph of the policy alone cannot establish identification. This distinction leaves the structural-estimation motivation intact while identifying the additional objects it would require.

5. CONTRACT CHANGES AND CERTIFIED POLICY REUSE

5.1. Operating Benefits and the Outside Option

The boundary contract permits a precise economic question: how do portfolio and preference choices respond when the benefit of remaining in the operating regime changes relative to the outside option? Add a constant flow benefit m to the running objective and a constant ℓ to the liquidation payoff, holding the diffusion, feasible controls, stopping rule, and consumption units fixed. These are changes in contract primitives. They need not be changes in the representation of a fixed preference ordering.

For a policy p , let B_n^p be its expected baseline utility, including G but excluding the adjustment charge. In addition to its discounted effort C_n^p , define

$$A_n^p = \mathbb{E}^p \int_{t_n}^{\tau} e^{-\rho(v-t_n)} dv, \quad Z_n^p = \mathbb{E}^p e^{-\rho(\tau-t_n)}. \quad (34)$$

All four functions depend on the initial state. The identity $\rho A_n^p + Z_n^p = 1$ holds pathwise before taking expectations. Thus the same policy's payoff is

$$J_n^p(m, k, \ell) = B_n^p + mA_n^p - kC_n^p + \ell Z_n^p = \ell + B_n^p + dA_n^p - kC_n^p, \quad d = m - \rho\ell. \quad (35)$$

The interpretation of d is the flow benefit of operation net of the annuity value of the liquidation increment. This is a comparison of two explicitly priced components of the contract. It does not require a preference-dependent normalization of consumption utility.

PROPOSITION 6—Contract contrast and economic moments: *Suppose the feasible policy class, transition law, and stopping rule do not depend directly on (m, k, ℓ) , and the required values are finite. Then*

$$V_n(m, k, \ell) = \ell + W_n(d, k), \quad W_n(d, k) = \sup_p \{B_n^p + dA_n^p - kC_n^p\}. \quad (36)$$

A joint change $(m, \ell) \mapsto (m + \rho h, \ell + h)$ leaves the entire optimal-policy correspondence unchanged and adds h to value. The function W_n is convex in (d, k) , increasing in d , and decreasing in k . If p_i has value loss at most e_i at $q_i = (d_i, k_i)$, then

$$(d_2 - d_1)(A_n^{p_2} - A_n^{p_1}) - (k_2 - k_1)(C_n^{p_2} - C_n^{p_1}) \geq -(e_1 + e_2). \quad (37)$$

At points of differentiability, optimal duration and effort equal $\partial_d W_n$ and $-\partial_k W_n$, respectively.

PROOF: Equation (35) proves the first claim for each policy before optimization. The supremum of the displayed affine functions is convex; their slopes in d and k have the stated signs. Approximate optimality gives $J^{p_1}(q_1) \geq J^{p_2}(q_1) - e_1$ and $J^{p_2}(q_2) \geq J^{p_1}(q_2) - e_2$. Adding and cancelling the intercepts proves (37). The differentiability statement follows from the supporting affine function of any optimizer. *Q.E.D.*

Two implications sharpen the interpretation of the portfolio exercise. First, increasing running utility alone is not a harmless normalization: it changes d and therefore the reward for discounted operating duration. Increasing the terminal payoff by the corresponding annuity value leaves d unchanged and cannot cause a portfolio reversal. Second, observations of choices and state paths alone cannot separately identify m and ℓ along this equivalence direction. This is an exact restriction on the optimal-policy correspondence, not a claim that d itself is identified. Identification of d still requires an observation model and economically informative variation. If the distribution of the termination time were independent of the policy, A^p would be common to all policies and even d would not change their rankings. Policy-dependent termination is therefore essential to the operating-benefit mechanism.

The supersolution argument for G in the baseline economy is not imposed on every changed contract: adding (m, ℓ) changes its flow-generator comparison by d . The

identities above concern payoff functionals and do not require a classical boundary trace for every counterfactual.

The coefficient-weighted charge is kC^p , whereas C^p is an unweighted quantity of adjustment effort. Neither Proposition 6 nor Theorem 5 makes kC^{p_k} monotone. We report both objects. We also distinguish a change in d from a change in correlation: the former changes the relative reward for remaining in operation, while the latter changes the state covariance and need not preserve the same policy-value features.

5.2. Value Envelopes on a Contract Interval

We next turn the economic decomposition into a computable decision rule. Consider a specified finite-horizon Markov model with positive continuation kernels. Its reward, including stopped payments, is affine in $q = (d, k)$; its action sets and transition kernels are fixed across q . A stored policy p_i has independently evaluated features (B^i, A^i, C^i) and hence value

$$J_n^i(s; q) = B_n^i(s) + dA_n^i(s) - kC_n^i(s). \quad (38)$$

Neural actors can supply such policies, but their trained critics are not substituted for these evaluated features.

Let $q_1, \dots, q_M$ be anchor contracts. Suppose $\bar{V}_n^i$ bounds the optimal value above at q_i . For $q = \sum_i \lambda_i q_i$, $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, define

$$L_n(s; q) = \max_i J_n^i(s; q), \quad U_n(s; q) = \sum_i \lambda_i \bar{V}_n^i(s). \quad (39)$$

The weights may be selected to minimize the upper bound; in the one-dimensional contract experiment, adjacent anchors give a chord. The stored policy need not be optimal at its anchor for the lower bound to be valid. The upper bound, in contrast, requires genuine anchor information: an evaluated feasible policy by itself is a lower bound, not an upper one.

THEOREM 7—Uniformly certified policy reuse: *Under the preceding hypotheses, $L_n(s; q) \leq V_n^*(s; q) \leq U_n(s; q)$. Define a feasible switching policy by*

$$i_n(s; q) \in \arg \max_i J_n^i(s; q), \quad p_n^{\text{sw}}(s; q) = p_{i_n(s; q), n}(s). \quad (40)$$

Then, at every date and state,

$$0 \leq V_n^*(s; q) - J_n^{\text{sw}}(s; q) \leq U_n(s; q) - L_n(s; q). \quad (41)$$

If the right-hand side is at most ε on a parameter set, the same policy-construction rule is ε -optimal throughout that set. There is no additional horizon multiplier on this independently evaluated value envelope.

PROOF: Every J^i is feasible, giving the lower bound. For any fixed policy p , affinity gives $J^p(q) = \sum_i \lambda_i J^p(q_i) \leq \sum_i \lambda_i \bar{V}^i = U(q)$. Taking the supremum proves the upper bound. For switching, the terminal equality is common to all stored policies. If $J_{n+1}^{\text{sw}} \geq L_{n+1}$, positivity of the continuation kernel implies

$$J_n^{\text{sw}} = R_n^{p_{in}} + K_n^{p_{in}} J_{n+1}^{\text{sw}} \geq R_n^{p_{in}} + K_n^{p_{in}} J_{n+1}^{in} = J_n^{in} = L_n.$$

Backward induction and the upper bound finish the proof.

Q.E.D.

The lower-envelope and switching arguments are established ideas in policy reuse; see Barreto et al. (2017), Chang (2021). The upper envelope is a convex-combination specialization of the policy-cache bound of Nemecek and Parr (2021). Our use of these results is explicit: the stopped-contract identity identifies the economically relevant parameter, the interval construction below certifies every parameter rather than selected queries, and the action-separation result translates value information into portfolio conclusions. Neither successor features nor policy caching is claimed as a new neural architecture.

The approximation theorem in Section 3 supplies one route to the required anchor upper bounds. If a feasible anchor policy has certified loss e_i , then $J^i(q_i) + e_i$ is an admissible $\bar{V}^i$. For the finite model executed here, backward maximization over an explicit action menu provides the anchor values and an independent feature recursion checks them. The target is this declared finite model: its operator discrepancy is zero relative to itself, not relative to Brownian first passage or to a continuum of actions. The latter approximation questions remain separate.

For a scalar interval $[a, b]$ at fixed k , write the two endpoint-policy lines as $b_a(s) + da_a(s)$ and $b_b(s) + da_b(s)$. Their maximum is a feasible lower envelope. The upper chord minus this maximum is concave and piecewise affine in d . Its maximum occurs at an endpoint or at the intersection

$$d_{\times}(s) = \frac{b_b(s) - b_a(s)}{a_a(s) - a_b(s)}, \quad (42)$$

when that intersection is defined and lies in $[a, b]$. The program checks these locations at every date and state. It inserts an anchor at a worst certified parameter and repeats until the maximum gap is below the stated tolerance. A denominator close to zero is handled as two effectively parallel lines, with an explicit numerical tolerance. This is a finite calculation of a continuum bound, not a dense parameter grid relabeled as uniform coverage. Supplement S.7 gives the error version and the numerical slack convention.

5.3. Certifying Economic Decisions, Not Just Values

A welfare bound need not determine a portfolio sign when two actions have nearly equal values. We therefore use a separate separation calculation. Partition the feasible first actions into a desired set A^+ and its complement A^- . For example, A^+ may contain strictly positive portfolio shares and A^- all nonpositive shares. Fix one action $a \in A^+$ and one stored continuation policy p_i . Its one-step value followed by that policy is affine in d :

$$Q_n^i(s, a; d) = R_n(s, a; d) + K_n^a J_{n+1}^i(d). \quad (43)$$

Let $\bar{Q}_n(s, b; d) = R_n(s, b; d) + K_n^b U_{n+1}(d)$, using the same upper chord throughout a contract interval.

PROPOSITION 8—Action separation over an interval: *If the same pair (a, p_i) satisfies*

$$\min_{d \in \{a_0, b_0\}} \left\{ Q_n^i(s, a; d) - \sup_{b \in A^-} \bar{Q}_n(s, b; d) \right\} > 0, \quad (44)$$

then every optimal first action belongs to A^+ for every $d \in [a_0, b_0]$.

PROOF: For every fixed $b \in A^-$, the difference between the feasible tail value and its upper continuation bound is affine on the interval. Positivity at both endpoints therefore implies positivity throughout. Taking the supremum over A^- preserves the strict separation, while $Q^i(s, a; d) \leq Q^*(s, a; d)$ and $Q^*(s, b; d) \leq \bar{Q}(s, b; d)$. No action in A^- can maximize the true finite-model criterion. *Q.E.D.*

The same desired action and continuation policy must be used at both endpoints of a test; selecting unrelated favorable endpoint actions would not prove (44). The implementation enumerates the feasible pairs and records the maximizing pair, both endpoint comparisons, and the resulting interval margin. An unresolved interval is distinguished from a zero effect. This permits a sharp portfolio conclusion away from a switching region while retaining an explicit uncertainty region where the sign is not separated.

6. COMPUTATION AND INDEPENDENT POLICY EVALUATION

6.1. The Stochastic Transition and Its Boundary

For the portfolio economy we use four equiprobable weak-Euler branches. Independent signs $z_1, z_2 \in \{-1, 1\}$ generate

$$\Delta u = \theta h + \sigma_u \sqrt{h} z_1, \quad \Delta X = [(r + \pi(\mu - r))X - c]h + \pi X \sigma_X \sqrt{h} \{ \varrho z_1 + \sqrt{1 - \varrho^2} z_2 \}. \quad (45)$$

Thus both variances and the cross covariance are present. On each branch we follow the line segment from s to $s + \Delta s$ until its first boundary intersection, at fraction $\alpha \in [0, 1]$ of the step. The branch contribution is

$$r_k(s, a) \frac{1 - e^{-\rho \alpha h}}{\rho} + e^{-\rho \alpha h} \{ 1_{\{\alpha=1\}} v_{n+1}(s + \Delta s) + 1_{\{\alpha<1\}} G(s + \alpha \Delta s) \}. \quad (46)$$

An initial boundary state stops immediately and receives G . An interior step that hits the boundary exactly may equivalently use its boundary datum. No branch is returned to an interior state after an exit.

This polygonal stopped random walk is an approximation to Brownian stopping, not an exact Brownian first-passage simulation. The reference operator uses positive bilinear interpolation for live continuations. The neural training operator evaluates its frozen continuation network at those same physical successor locations. Policy evaluation subsequently uses a separate NumPy implementation with positive interpolation. These two approximations are compared rather than assumed to be identical.

Generator tests use $1, u, X, u^2, X^2, uX$. At an interior test point, their discrepancies have the expected drift-squared Euler remainder and recover the complete covariance matrix. A separate test verifies that opposite preference controls produce different conditional means even when the displacement is smaller than a grid cell. Interpolation preserves that difference; nearest-node rounding can erase it. The differential actor graph, finite-step actor graph, and immediate stopping boundary are also tested independently.

Table I reports a refinement sequence with $\max(\Delta u, \Delta X)^2/h$ decreasing from 0.04 to 0.01. This matters because interpolation error divided by the time step must be controlled in a generator approximation. The action mesh is held fixed in this sequence,

TABLE I
JOINT STATE-TIME REFINEMENT OF THE STOPPED PREFERENCE MODEL

Grid	Dates	V_0	C_0	Seconds
17×25	4	-1.140548	0.009344	0.31
33×49	8	-1.136916	0.010335	2.07
65×97	16	-1.135019	0.010902	15.45

Initial state $(2, 1.25)$; $k = 2$. All rows optimize $5 \times 5 \times 7$ consumption, adjustment, and portfolio actions. C_0 is discounted integrated $\theta^2/2$. These are finite-approximation values, not certified continuous-time errors.

TABLE II
RE-SOLVED PREFERENCE-ADJUSTMENT COST PANEL

k	V_0	C_0	c_0	θ_0	π_0
0.5	-1.117524	0.014614	0.800	0.200	-0.500
2.0	-1.135835	0.009888	0.800	0.200	-0.500
8.0	-1.160291	0.001432	0.800	0.100	-0.500

Each row re-solves all dates and all three control coordinates on the same 25×37 grid, eight dates, and $9 \times 9 \times 13$ action mesh. The initial state is $(2, 1.25)$. The comparison concerns cumulative effort, not a universal pointwise policy ordering.

and action refinement is assessed separately. These calculations diagnose discretization sensitivity; three meshes do not themselves constitute an error certificate for the continuous-time value.

6.2. Preference-Cost Comparisons

The cost panel solves the entire finite problem again at $k = 0.5, 2, 8$, using a 25×37 state grid, eight dates, and $9 \times 9 \times 13$ controls. Consumption, preference adjustment, and portfolio share are all optimized. Discounted effort is evaluated under the resulting policy, including its effect on the exit time. Table II reports the initial state $(u, X) = (2, 1.25)$.

Discounted cumulative effort falls as k increases, consistently with Theorem 5. The portfolio and consumption coordinates at this initial state are at their bounds for all three displayed coefficients. This is not evidence for the interior formulas in (29). In particular, a short portfolio position in this calibration should not be described as an unconditional preference-hedging effect. The liquidation contract, the level of the running utility, and the portfolio's effect on exit jointly determine the optimum. Occupation-weighted exit probabilities are computed by forward propagation of the interpolation-defined Markov chain; an unweighted fraction of exiting grid branches would measure a different object.

6.3. Neural Policies in the Preference Economy

The ordinary neural proposal has two tanh hidden layers of width 32 and three bounded outputs. The safeguarded critic has two tanh hidden layers of width 64. Its output is $G(s) + (1 - t)m(s)N_{\xi}(s)$, where m vanishes on every face of the rectangle. The implementation uses separate L-BFGS fits, 2,048 uniformly sampled training states

TABLE III
NEURAL POLICIES AND INDEPENDENT EVALUATION IN THE STOPPED ECONOMY

Run	$V_R - J_0$	Critic RMSE	Critic max.	Gain max.	Seconds
Pilot 101	0.026720	0.066694	0.676974	1.340226	13.5
Safe 101	0.001018	0.047925	0.298592	0.228318	88.3
Safe 202	0.001366	0.044197	0.346920	0.210235	83.0
Safe 303	0.001177	0.052661	0.934824	0.228568	81.5

V_R is the separately solved 33×49 , eight-date, $9 \times 9 \times 13$ reference at (2,1.25). Critic errors compare the initial critic with its independently evaluated policy over the whole grid. Gain is the largest feasible one-step improvement over all interior states and dates against the evaluated continuation. The pilot has a smaller critic and training budget; it is not a matched ablation. Hard boundary discrepancies are evaluated separately in the raw record.

TABLE IV
FULL-VECTOR POLICY DISCREPANCIES AGAINST THE MATCHED REFERENCE

Run	MAE c	MAE θ	MAE π	Max. c	Max. θ	Max. π
Pilot 101	0.0136	0.0678	0.3212	0.562	0.400	1.300
Safe 101	0.0047	0.0165	0.1065	0.562	0.400	1.300
Safe 202	0.0047	0.0157	0.0954	0.562	0.400	1.300
Safe 303	0.0048	0.0158	0.1085	0.562	0.400	1.300

Mean absolute and maximum discrepancies include every interior common node and all eight dates. Coordinates have different economic units and are not added together or called a Hamiltonian gain. Both policies use their stated finite approximations; these are not errors against a continuous-time oracle.

per seed, eight dates, and a feasible $5 \times 5 \times 7$ comparison set containing the actor’s proposal. No grid reference values or optimal reference policies enter these training targets.

After training, the entire feedback sequence is frozen and evaluated on a 33×49 grid. The comparison reference uses the same number of dates and a finer $9 \times 9 \times 13$ action mesh. A separate $7 \times 7 \times 11$ deviation search uses the *evaluated policy continuation*, not the learned critic, and includes the incumbent action. Consequently, a reported gain is a feasible payoff improvement. It is not a norm of the difference between two actions and is not an upper bound over the continuum of actions.

Table III reports all three safeguarded seeds and retains a lower-budget, unsafeguarded pilot. The pilot differs in width and optimization budget, so its comparison is a development diagnostic, not an isolated causal ablation. Table IV reports errors for the full control vector. A small error at the initial state does not imply a small maximum error near the boundary, and a policy coordinate can change substantially near a switch. The critic, policy value, and feasible-deviation columns therefore remain distinct.

The hard boundary architecture attains the displayed boundary condition to numerical precision. It does not certify the interior value or the Brownian stopping approximation. The sizable worst-case diagnostics identify where additional computation is needed for a uniform guarantee; they are not suppressed by averaging across states or seeds. Conversely, the initial-state comparison and the re-solved cost panel provide economic information that is not captured by the largest training residual alone.

TABLE V
NEURAL-POLICY LOSS BEFORE EVALUATED-CONTINUATION REPAIR

Seed	Feasible gain	Initial loss	Population loss	All-node loss
101	0.228318	0.001058	0.000730	0.236122
202	0.215071	0.001406	0.000783	0.245815
303	0.228568	0.001217	0.000800	0.236109

All losses are value differences in the same finite economy, relative to exhaustive backward optimization. The action menu is the union of the reference and original diagnostic meshes and three frozen neural proposals at each node. The initial state is $(2, 1.25)$; the population is uniform on grid nodes in $[1.6, 2.4] \times [0.8, 1.6]$. “All-node includes every date and state, not only occupied states. The repaired policy has zero remaining gain in the computed backups over this menu. The published diagnostic mesh is contained in the new menu.

6.4. Repair and Contract Counterfactuals

The policy errors above can be repaired without treating the learned critic as a value certificate. We freeze the three neural policy arrays and form, at each date and state, the union of the $9 \times 9 \times 13$ reference mesh, the $7 \times 7 \times 11$ diagnostic mesh, and the three frozen proposals. The common mesh contains 1565 distinct actions; the complete menu has 1568 entries per node. Including the original diagnostic mesh is important: two different action meshes need not be nested merely because one has more points. Every previously tested profitable deviation is feasible in the repair calculation.

The target is now an explicit eight-date, 33×49 killed Markov chain. A branch’s live interpolation weights are nonnegative; its stopped payoff, discounted duration, and adjustment effort are immediate rewards. Backward maximization uses the *evaluated* next-date optimal continuation, and a separate fixed-policy recursion evaluates (B, A, C) . In this finite model, the incumbent proposals are feasible and the repaired policy weakly improves their values at every node. Table V gives the pre-repair errors. The post-repair Bellman gain over the entire stated menu, using independently evaluated policy continuations, is 4.441×10^{-16} , below 10^{-10} .

The largest initial-fit gains occur near the low-risk-aversion and high-wealth portion of the rectangle. Their exact dates, states, and occupation masses are saved. Under both the focal state and the stated initial population, the single worst-gain nodes have zero occupation mass. This explains why a large all-state error can coexist with a much smaller initial welfare loss; it does not make the global error disappear. The population losses and the occupation-weighted gains are distinct calculations, and the latter are not used as upper bounds on optimal-policy loss. The repair addresses all states rather than discarding the low-occupation ones.

At $k = 2$, the adaptive construction uses 47 anchors to cover every $d \in [0, 1]$. Its maximum two-policy chord gap is 0.00093070, below 10^{-3} at every date and state. The library evaluation does not retrain an actor or solve a new control maximization at a counterfactual query. A direct comparison at 101 parameter values finds a maximum value loss of 2.409×10^{-4} ; this comparison validates the implementation but is not the source of the continuum claim. The latter follows from the complete breakpoint enumeration in Section 5.

Table VI reports five independently optimized initial-state counterfactuals. Applying Proposition 8 to the full menu certifies a negative optimal initial share on $[0, 0.361017]$ and a positive share on $[0.373654, 1]$, with inward-rounded displayed endpoints. The intervening band is unresolved by this certificate. The sign conclusion uses action-value

TABLE VI
OPERATING-BENEFIT COUNTERFACTUALS

d	Value	Duration	c_0	θ_0	π_0	Sign value gap
0.00	-1.136507	0.887826	0.80	0.20	-0.50	-0.000876
0.25	-0.914548	0.887930	0.80	0.20	-0.50	-0.000277
0.50	-0.692266	0.890233	0.80	0.20	0.80	0.000299
0.75	-0.469703	0.890536	0.80	0.20	0.80	0.000869
1.00	-0.246682	0.894175	0.80	0.20	0.80	0.001281

Initial state $(u, X) = (2, 1.25)$, $k = 2$. Duration is A^p , not the probability of survival. The final column is the best value among nonnegative-share first actions minus the best value among nonpositive-share first actions, using the optimal finite-model continuation. A positive entry favors the positive-share group. These five direct solves are distinct from the continuum certificate, which uses interval envelopes and fixed feasible action–continuation pairs.

separation and therefore does not confuse a 10^{-3} global welfare tolerance with a 10^{-3} tolerance on an action coordinate.

The contract mechanism is economically specific. Raising the operating benefit rewards longer discounted operation and changes the portfolio from a short position toward a long one in the displayed economy. Adding the corresponding annuity amount to liquidation leaves the full policy correspondence unchanged; an independently evaluated terminal-exposure recursion verifies the accounting identity. The endpoint reversal also occurs when the shock correlation is set to zero or to 0.25. It is therefore not an unconditional sign prediction from negative preference correlation. The model does not claim that a unit operating benefit is empirically estimated or that this portfolio response holds under every liquidation contract.

Supplement S.8 reports joint state, time, and action refinements. The two endpoint signs persist, while the portfolio at $d = 0.5$ changes from negative on the coarsest economy to positive on the two finer ones. Value levels also move with resolution. This makes the location of a diffusion-model switch a separate approximation question; it does not invalidate the all-parameter statement for the specified finite target. The original continuous-time operator and its error-discrepancy term remain in the analysis.

For an actual 101-query workload, full direct solutions take 184.86 seconds, compared with 1.43 seconds for library selection and complete policy evaluation. Building the library and common kernel takes 87.47 seconds, including the interval-certificate calculation. These are incremental times given the same frozen proposals to both methods. The experiment measures a reusable family of complete policies and moments, rather than extrapolating from a single network forward pass. The stored library contains finite-state policy values; its construction does not establish a neural compression advantage in this low-dimensional economy.

6.5. Coupled Stochastic Resource Allocation

To examine a genuinely dynamic shared constraint, let $x_n \in \mathbb{R}^d$ denote productive stocks and $a_n \in \mathbb{R}_+^d$ investment. The three-period model is

$$x_{n+1} = A_d x_n + a_n + e_{n+1}, \quad A_d = 0.8I + 0.04(I_{+1} + I_{-1}), \quad \sum_i a_{ni} \leq 0.25d. \quad (47)$$

Here L_{+1}, L_{-1} denote the adjacent off-diagonal matrices. The shock has four equally likely realizations, $e = z_1 \ell_1 + z_2 \ell_2$, where $\ell_{1i} = 0.06$ and ℓ_{2i} is evenly spaced from -0.04 to 0.04 . Its covariance is rank two and generally has nonzero cross-stock entries. Heterogeneous initial stocks, neighbor spillovers, a common budget, and the aggregate cost couple the state coordinates.

The objective is expected total cost,

$$C^\pi(x) = \mathbb{E}_x^\pi \left[\sum_{n=0}^2 \left\{ q(x_n) + \frac{0.1}{d} \|a_n\|^2 \right\} + q(x_3) \right], \quad (48)$$

$$q(x) = \frac{\|x - \mathbf{1}\|^2}{d} + 0.2 \left(\frac{\mathbf{1}^\top (x - \mathbf{1})}{d} \right)^2.$$

This is a finite stochastic economy, not a discretization claim about a different diffusion. All 64 terminal shock paths are used when a policy is evaluated. The independent reference optimizes a nonanticipative scenario tree with 21 decision nodes per initial state. Its objective is convex quadratic and its nodewise resource constraints are genuine restrictions on investment.

The cost critic is

$$\hat{C}_n(x) = q(x) + b_n + \ell_n^\top (x - \mathbf{1}) + \sum_{j=1}^J w_{nj} [h_j^\top (x - \mathbf{1}) + b_j]^2, \quad w_{nj} \geq 0. \quad (49)$$

The fixed hidden features include coordinate, aggregate, and random directions; $J = 6d + 102$. Output coefficients are fitted by constrained least squares to the separated Bellman targets. A width-32 tanh proposal network uses a slack-softmax feasibility map and is trained against the improved controls. At deployment, projected convex improvement uses the analytic critic gradient, checked against automatic differentiation, and the gap (15). Squared-ReLU features are suitable here because the finite-step operator needs a value and an action gradient, not a classical second-derivative residual.

For each of $d = 4, 8, 16$ and seeds 101, 202, and 303, training uses 1,024 draws from $[-0.25, 1.65]^d$. Evaluation uses 32 fresh initial states from $[0, 1.2]^d$ and their complete shock trees. The reference is solved both to the common cost tolerance 10^{-3} for timing and to a gap below 10^{-7} for independent validation. Formula (16) gives a numerical upper bound on each tested policy's cost loss. These are double-precision convexity certificates; recorded feasibility and gradient discrepancies accompany them, rather than an assertion of interval-arithmetic proof.

Every displayed run meets the common cost-error target at its tested initial states. The shared constraint binds frequently, including at most or all initial states depending on dimension and seed. Thus the experiment tests a dynamically consequential budget rather than an inactive constraint. The same trained proposal networks are also evaluated without the deployment safeguard. This matched ablation, reported in the supplement, isolates the effect of the improvement calculation without changing network weights or training data.

At this query batch and tolerance, the specialized convex reference is faster end to end than training and evaluating the neural representation. The latter delivers a reusable feedback approximation, but a crossover in amortized cost requires a separate measurement. The experiment establishes accurate constrained neural control for this model and a transparent cost-accuracy comparison; it does not establish superiority over a solver that exploits the model's convex structure.

TABLE VII
COUPLED STOCHASTIC RESOURCE ALLOCATION AT A COMMON ACCURACY TARGET

d	NBO upper loss	Proposal only	NBO sec.	Reference sec.	Passed
4	0.000645	0.01190	1.737	0.172	3/3
8	0.000461	0.09541	2.241	0.190	3/3
16	0.000567	0.06807	3.180	0.226	3/3

Loss columns are worst held-out cost-loss upper bounds across three seeds and 32 initial states per seed. Proposal only removes deployment improvement from the same fitted actor weights. Timing is the median across seeds at a 10^{-3} cost-loss target; NBO includes fitting and full evaluation, and the reference solves the same initial-state batch to its 10^{-3} gap target. A separate 10^{-7} reference gap is used for auditing, not for the timing comparison. The reference is a convex nonanticipative scenario tree, not a sparse grid.

TABLE VIII
REPRESENTATION AND ACTOR-FREE COMPARISONS

d	Actor-start loss	Zero-start loss	Quadratic loss range	Actor-zero difference
4	6.446×10^{-4}	6.446×10^{-4}	2.065×10^{-3} – 3.203×10^{-3}	1.256×10^{-10}
8	4.607×10^{-4}	4.607×10^{-4}	1.668×10^{-3} – 6.692×10^{-3}	1.104×10^{-10}
16	5.665×10^{-4}	5.665×10^{-4}	5.444×10^{-3} – 6.777×10^{-3}	2.679×10^{-11}

Each dimension includes seeds 101, 202, and 303 and the same 32 initial states per seed. The first two columns give the largest certified lifetime cost-loss upper bound across those seeds. The quadratic column spans the three maximum losses of a separately trained full positive-semidefinite quadratic critic. All use the same feasible controls and full shock trees; a reference Frank–Wolfe gap below 10^{-7} is added to each measured cost difference. Actor and zero starts use identical frozen nonlinear critic weights.

6.6. What the Actor and the Continuation Representation Contribute

The actor does not enter the resource experiment’s critic targets: the target controls are computed before actor distillation. A same-weight comparison must therefore separate the value of continuation approximation from the value of actor initialization. We reload all nine checkpoints and solve the same convex improvements from either the actor output or zero. The largest complete-policy cost difference between these two arms is 1.256×10^{-10} . Both attain the stated 10^{-3} cost tolerance on every original held-out state. Actor initialization can save iterations, but those savings do not by themselves imply that its training cost is amortized.

The nonneural comparator is a full convex quadratic, $q(x) + b + \ell^\top(x - \mathbf{1}) + (x - \mathbf{1})^\top H(x - \mathbf{1})$ with $H \succeq 0$. It is fitted by projected least squares on the same training-state distribution and used in its own backward recursion. Unlike a diagonal quadratic, it permits every cross-state term. Its projected-gradient stopping diagnostic and Hessian eigenvalues are recorded. Table VIII shows that the richer squared-ReLU continuation class attains the common welfare target whereas these quadratic fits do not. This is evidence for the representational flexibility used here, not a theorem that a neural representation dominates all nonneural approximation classes.

We next evaluate a fresh sequence of initial states at query volumes 32, 128, 512, and 2,048. No initial state is added to the training set. Each query includes the complete endogenous policy tree and its 64 terminal shock paths. Table IX charges for critic training, zero-start online improvement, and policy evaluation, and compares that sum with the specialized scenario-tree solver at the same 10^{-3} cost tolerance. The independent tighter reference remains solely an accuracy check for both methods. At

TABLE IX
FRESH-STATE QUERY WORKLOADS AT A COMMON COST TOLERANCE

Queries	Max. loss bound	Training	Query	Reference	Cost ratio
32	6.024×10^{-4}	0.812	0.040	0.152	0.18
128	6.574×10^{-4}	0.812	0.119	0.420	0.45
512	6.574×10^{-4}	0.812	0.600	1.477	1.05
2048	6.754×10^{-4}	0.812	2.600	5.708	1.67

Dimension four, seed 101; held-out state seed 541021, uniform on $[0, 1.2]^4$. Larger workloads extend the same fresh-state sequence. Times are seconds. Query time includes zero-start convex improvement and all 64 terminal paths per initial state. Reference time is for independent nonanticipative optimization to the common 10^{-3} cost tolerance. The cost ratio is reference time divided by critic training plus query time. Each reported maximum policy-loss bound is below 10^{-3} . Timings are medians of three evaluations; the separate tighter validation reference is not charged to either competing solver.

the largest workload the measured end-to-end cost ratio is 1.67, with a maximum policy-loss bound of 6.754×10^{-4} . The intermediate workloads show the crossover rather than assuming it.

The useful reusable object in this comparison is the nonlinear continuation approximation. The actor’s separate distillation time, initialization time, optimizer iterations, and projected break-even query counts are reported in Supplement S.8. The break-even projection is labeled as such; only the fresh-state workload table reports an observed comparison. The conclusion is consequently conditional on the model, tolerance, query distribution, and computing environment. It does not rely on comparison with an infeasible tensor grid or on treating a three-period model as a high-dimensional complexity theorem.

6.7. Smooth and High-Dimensional Controls

An additional stochastic LQR experiment uses correlated Gaussian shocks, neighbor coupling in the transition, and a rank-one cross-state cost at dimensions 4, 8, and 16. Its critic has quadratic features, its actor is linear, and its independent reference is Riccati recursion. This structured control checks backward evaluation, covariance contributions, actor coefficients, and full-policy values. It is not an experiment with a generic multilayer critic, nor a substitute for the constrained resource application.

Computational cost depends on the operator and representation. For a differential critic and rank- r loading B , the exact trace can be evaluated using r directional Hessian contractions. A K -probe trace instead uses K such contractions after forming the probe directions. Their total costs include network differentiation, loading multiplication, action optimization, sampling, and the iterations needed to achieve an error target. Per-iteration differentiation does not establish a polynomial solution-complexity theorem, and no threshold such as dimension 50 follows from that calculation. The semigroup experiments report their own costs without attributing them to unexecuted stochastic-Hessian acceleration.

TABLE X
EXECUTED HOMOTHETIC POLICY EVALUATION AND IMPROVEMENT

Model	Learned m	Learned π	Policy max. error	Relative A error
Merton	0.04124996	0.74999998	3.94e-08	9.13e-13
Epstein–Zin	0.04550010	0.30000000	1.03e-07	6.89e-12

The coefficient $A = \exp(q)$ and bounded actor parameters are learned by separated updates. Analytical values are used only to compute the displayed errors. The sign condition and normalized HJB are tested separately. These runs use homothetic features rather than an unrestricted neural value class.

7. RECURSIVE UTILITY, TEMPORAL SELVES, AND DYNAMIC GAMES

7.1. Merton and Epstein–Zin Policy Evaluation

The smooth portfolio benchmark uses stationary infinite-horizon CRRA utility with $\rho = 0.04$, $\gamma = 2$, $\mu = 0.08$, $r = 0.02$, and $\sigma = 0.2$. With $c = mX$ and no terminal bequest, the optimal ratios are

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2} = 0.75, \quad m^* = \frac{\rho + (\gamma - 1)\{r + (\mu - r)^2 / (2\gamma\sigma^2)\}}{\gamma} = 0.04125. \quad (50)$$

These are stationary quantities, not finite-horizon policies with an arbitrary terminal condition. Under the displayed optimum, the discounted negative wealth moment in the verification argument decays at rate $m^* > 0$.

For recursive utility, put $a = 1 - 1/\psi$ and $b = 1 - \gamma$. The difference-form Epstein–Zin driver on $c > 0$, $bV > 0$ is

$$f_{EZ}(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}. \quad (51)$$

The logarithmic case requires a limit rather than substitution at $a = 0$. In the numerical regime $\gamma = 5$, $\psi = 1.5$, with the same asset and discount parameters, homothetic optimization gives

$$\pi^* = 0.3, \quad m^* = \psi\rho + (1 - \psi)\{r + (\mu - r)^2 / (2\gamma\sigma^2)\} = 0.0455. \quad (52)$$

For $V = AX^b/b$, the coefficient is $A = (\rho(m^*)^{a-1})^{b/a}$. Consumption utility is increasing on the declared domain. A negative value when $b < 0$ is required by that domain and is not a sign error.

We execute fixed-policy evaluation for the unknown coefficient A and separate automatic-differentiation updates for bounded consumption and portfolio parameters. The coefficient is represented by $A = \exp(q)$; it and the controls are learned, not set to their targets. The analytical ratios are used afterward for validation. Table X distinguishes these homothetic-feature solves from the ordinary neural networks used for endogenous preferences.

Supplement S.4 derives the policy equations, the positive branch of the evaluation equation, and a verification class with the required transversality. For the displayed recursive optimum, the linearized discount rate is 0.1115, the relevant wealth moment grows at rate 0.066, and their difference is $0.0455 > 0$. A separate finite-horizon architecture test imposes $bV > 0$ and an exact terminal sign by a transformed output. It is a domain test, not an additional solution of a finite-horizon Epstein–Zin problem.

TABLE XI

SOPHISTICATED CONSUMPTION RATIOS AND INDEPENDENT DEVIATIONS

β	c_0/w	c_1/w	c_2/w	c_3/w	Eval. residual
0.7	0.2827991	0.3401848	0.4312698	0.5978885	1.78e-15
1.0	0.2163112	0.2651939	0.3467520	0.5099987	1.78e-15

Four consumption dates and terminal log wealth, $\Delta = 1$, $\rho = 0.04$. Continuation is evaluated without multiplying by β again. Independent bounded one-shot optimizations against these frozen continuation values give no gain exceeding 10^{-9} on the tested wealth levels.

The role of these calculations is to test recursive value dependence and the evaluation graph. Neither a low residual nor the positivity transform is by itself a proof of recursive-utility existence for every policy. Equally, the fact that an argmax has no analytic formula does not define a fully nonlinear PDE. Semilinearity concerns dependence on the highest-order derivatives. Control of diffusion can make the HJB fully nonlinear, whereas a difficult drift-control maximization with a fixed diffusion need not do so. The FBSDE literature also contains value representations rather than exclusively Pontryagin adjoints (Han, Jentzen, and E, 2018); these methods should not be excluded by that distinction.

7.2. Sophisticated Temporal Selves

Let the state obey $s_{n+1} = F(s_n, c_n, \varepsilon_{n+1})$ and let $\delta = e^{-\rho\Delta}$. A sophisticated current self evaluates a one-period deviation using $\beta\delta$ on ordinary continuation utility. The equilibrium system is

$$c_n^*(s) \in \arg \max_{c \in \mathcal{C}(s)} \{u(c)\Delta + \beta\delta \mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (53)$$

$$V_n(s) = u(c_n^*(s))\Delta + \delta \mathbb{E}[V_{n+1}(F(s, c_n^*(s), \varepsilon))], \quad V_N = G. \quad (54)$$

The factor β belongs in the current self's improvement criterion but not in each step of the ordinary continuation evaluation. Repeating it in (54) instead defines a different discount recursion. We use the same equations in the main text, supplement, and implementation.

The executed example has four consumption dates, terminal log wealth, no interest, $w_{n+1} = w_n - c_n$, $0 < c_n < w_n$, and $\Delta = 1$. With $V_n(w) = A_n \log w + B_n$ and $A_N = 1$, backward induction gives

$$z_n \equiv \frac{c_n}{w_n} = \frac{1}{1 + \beta\delta A_{n+1}}, \quad A_n = 1 + \delta A_{n+1}, \quad B_n = \log z_n + \delta \{A_{n+1} \log(1 - z_n) + B_{n+1}\}. \quad (55)$$

Strict concavity establishes the one-shot optimum. We compare it with independent bounded scalar optimization and evaluate the ordinary recursion at four wealth levels. All consumption columns are ratios, not levels. The no-present-bias case $\beta = 1$ recovers exponential discounting.

A two-consumption-date example with terminal log wealth and $\delta = 1$ makes the distinction transparent: the sophisticated first ratio is $1/(1 + 2\beta)$, whereas the erroneous geometrically repeated-beta recursion gives $1/(1 + \beta + \beta^2)$. They equal 0.416667 and 0.456621 at $\beta = 0.7$. The discrepancy is economic, not a change of notation.

7.3. Stochastic Capacity Competition

For player i , evaluation freezes the policy profile, and improvement maximizes only player i 's criterion with rivals' controls detached. A dynamic equilibrium must also survive unilateral changes in the player's entire continuation policy. We therefore distinguish a one-period Hamiltonian gap from dynamic exploitability,

$$\text{Exploit}_i(s) = \sup_{\tilde{\pi}_i} J_i(\tilde{\pi}_i, \pi_{-i}; s) - J_i(\pi_i, \pi_{-i}; s). \quad (56)$$

A renewal-capacity regression has two firms, binary capacities $K_i \in \{0, 1\}$, and demand $D \in \{1, 1.3\}$ with transition matrix $\begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{pmatrix}$. Output is $q_i = K_i/3$, and revenue is $q_i(D - q_i - q_j)$. Investment $a_i \in [0, 1]$ is the probability of capacity one next period, conditional on which the firms' capacity draws are independent. Its cost is $0.4a_i^2/2$. This includes the cost of maintaining capacity; it is not a free reset. The horizon is four dates, the discount factor is 0.95, and terminal value is $0.05K_i$.

For every date and all eight states, stage best responses are clipped affine functions of the rival's investment. Their composed contraction is checked, and iteration is repeated from three initial profiles. After the equilibrium profile is computed, a separate backward best-response program freezes the rival at every date and optimizes the deviating firm's continuation. The original output compared the final, date-zero best-response value. We now compare the best-response value with the equilibrium value inside every step of the backward recursion; the all-date gain is below the stated 10^{-10} tolerance. The saved policies and values cover the complete finite state space.

The renewal transition makes next capacity independent of current capacity, conditional on the chosen investment. Its early policy consequently has a closed form. Writing P for the demand transition matrix, the first three dates satisfy

$$a_n(D) = \frac{\delta \{\mathbb{E}[D' | D] / 3 - 1/9\}}{\kappa + \delta/9}, \quad a_3 = \frac{0.05\delta}{\kappa}. \quad (57)$$

We retain this case as a regression test and add economically persistent capital. With survival $s = 0.8$, the new twelve-date transition is

$$\Pr(K_{i,n+1} = 1 | K_{i,n}, a_{i,n}) = sK_{i,n} + (1 - sK_{i,n})a_{i,n}. \quad (58)$$

Investment creates capacity when none is installed and offsets the risk of losing an installed unit. Current capacity now changes the marginal return to investment. All other flow-profit, cost, demand-transition, and terminal-value primitives are held fixed.

The one-period best responses remain clipped affine functions of the rival's action. The implementation enumerates all nine pairs of binding and interior constraints, rather than imposing interiority. The maximum composed best-response slope product is 0.090398, below one. Table XII shows dependence on installed capacity, the rival's capacity, demand, and the remaining horizon. A separate full best-response recursion finds a maximum unilateral dynamic gain of 4.441×10^{-16} over all 192 date-player-state combinations. Deliberately replacing one late-date action by a nonoptimal investment produces a large detected gain at a recorded date and state, checking that the diagnostic does not inadvertently reduce to date zero.

This is an independent dynamic-game reference, not a trained high-dimensional game network. A separate automatic-differentiation test verifies that the own-action gradient is present and the rival gradient is absent. The static Cournot comparison,

TABLE XII
INVESTMENT WITH PERSISTENT CAPACITIES

K_1	K_2	$a_{1,0}(D=1)$	$a_{1,0}(D=1.3)$	$a_{1,10}(D=1)$	$a_{1,11}(D=1)$
0	0	0.700239	0.816869	0.521484	0.118750
0	1	0.665094	0.812634	0.442261	0.118750
1	0	0.141975	0.163600	0.108478	0.023750
1	1	0.133116	0.162538	0.088661	0.023750

Twelve investment dates and capacity survival $s = 0.8$. Capacity-one probability is $sK_i + (1 - sK_i)a_i$. Investment therefore depends on installed capacity and on continuation opportunities. The largest independently computed full unilateral deviation gain over all 192 date–player–state combinations is 4.441×10^{-16} .

with Nash quantities $1/3$ and joint-profit quantities $1/4$, remains a diagnostic for the distinction between equilibrium and collusion, not a substitute for the dynamic check. Supplement S.5 sets out the equilibrium operator, resource normalization for larger markets, and a conditional small-player approximation bound. Symmetry of initialization is not used as an equilibrium certificate.

The portfolio and recursive-utility benchmarks build on [Merton \(1971\)](#), [Epstein and Zin \(1989\)](#), and [Duffie and Epstein \(1992\)](#). The normalization and admissible comparison class used here are specified in Supplement S.4.

8. CONCLUSION

Neural Bellman Operators make policy proposal, evaluation, improvement, and certification separate economic computations. That separation is useful when it supports a decision: whether a stored policy remains adequate after a contract changes, whether a portfolio sign is determined by economic incentives rather than numerical error, and whether a reusable continuation approximation pays for its construction.

For the stopped preference economy, the relevant contract contrast is the operating benefit net of the liquidation annuity. Independently evaluated policy features and certified value envelopes provide a uniform welfare guarantee over its full displayed interval. Action-separation bounds identify negative and positive portfolio regions and retain an unresolved switching band. The finite implementation includes every previously diagnosed candidate deviation and repairs the initial neural policies at all dates and states. The exact normalization identity and the separate correlation comparisons identify the operating-duration mechanism without converting it into a universal hedging claim.

The resource experiment establishes why the continuation representation and the actor should be assessed separately. A nonlinear convex continuation fits the tested problem more accurately than a full convex quadratic. Reuse becomes useful at a measured workload, while the actor primarily changes initialization and must earn back its own training cost. The persistent-capacity reference, recursive-utility updates, and temporal-self calculations preserve the broader framework and test its economically distinct operators.

The continuous-time equations, finite transition models, trained proposals, and independent policy values remain distinct objects. This distinction makes the computational results stronger: each numerical guarantee states the policy class, boundary mechanism, parameter range, and error tolerance it actually covers. It also permits subsequent

extensions to change the approximation architecture without changing the economic meaning of the certificate.

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